DATA ANALYTICS PORTFOLIO · RETAIL PERFORMANCE STUDY

# Superstore: where revenue stopped meaning *profit*

A four-year analysis of 8,399 order lines across sales, margin, product mix, customers, discounting and fulfilment — built end-to-end in Python, PostgreSQL and Power BI.


PREPARED BY
**Abdullah Waseem**


PERIOD ANALYSED
**Jan 2009 – Dec 2012**

ORDER LINES
**8,399**

UNIQUE ORDERS
**5,496**

CUSTOMERS
**795**

ISSUED
**September 2026**

HOW TO READ THIS REPORT

1. **Every section follows the same shape.** A metric strip states the position, one or more exhibits show the evidence, and an *Evidence → Reading → Implication* block separates what the data says from what it means and what to do about it.
2. **Exhibits are numbered and self-contained.** Each carries its own note explaining what the chart is doing. Charts backed by a long series carry the full data table directly beneath them.
3. **Figures reconcile to one source.** Every number was recomputed directly from `superstore_cleaned.csv` for this report and cross-checked against the notebook and SQL outputs.
4. **Where the evidence is weak, the report says so.** Two findings are explicitly qualified rather than presented as conclusions — see §9 and §16. That is deliberate.

DOCUMENT CONTROL

| | |
|---|---|
| REPORT | **Superstore Performance Review** |
| AUTHOR | Abdullah Waseem |
| ISSUED | September 2026 |
| STATUS | FINAL |
| ANALYSIS PERIOD | 1 Jan 2009 – 30 Dec 2012 **(1,459 days)** |
| SOURCE | `superstore_cleaned.csv`<br>8,399 rows × 33 columns |
| GRAIN | One row = one order line |
| CURRENCY | USD, unrounded in source |
| METHODS | Python · PostgreSQL · Power BI |
| SCOPE | 39 business questions · 10 SQL analyses |
| AUDIENCE | Commercial & category management |

# 01 Executive summary

THE HEADLINE NUMBER IS HEALTHY. THE DISTRIBUTION UNDERNEATH IT IS NOT.

| REVENUE | PROFIT | MARGIN | AVG ORDER VALUE |
|---|---|---|---|
| **$14.92M** | **$1.52M** | **10.20%** | **$2,714** |
| 2009–2012 | net of $1.10M losses | blended, 4 years | 5,496 orders |

LOSS-MAKING LINES

**50.77%**

4,264 of 8,399

Superstore converts 10.20% of revenue into profit across the four-year window — a defensible headline. But that figure is a net result, not a description of the business. Slightly more than half of all order lines lose money, and the profitable half is large enough to cover them with $1.52M left over.

The same pattern repeats at every level of aggregation examined in this study — region, category, sub-category, product, customer and year. **Revenue rank and profit rank persistently disagree.** The West is the second-largest region by revenue and the worst by margin. Furniture sells nearly as much as Technology and returns 2.27% against Technology's 14.81%. The second-best-selling product in the catalogue loses money on every unit. In 2012 revenue rose 8.24% while profit fell 10.37%.

EVIDENCE

4,264 of 8,399 order lines carry negative profit, totalling **–$1,102,503**. Blended margin still lands at 10.20%.

READING

Losses are not concentrated in one region, segment or channel. They are distributed and structural — a pricing and product-mix problem, not a discounting or logistics one.

IMPLICATION

The lever with the largest expected return is margin discipline on demand the business already has, not incremental sales volume.

## WHAT THE ANALYSIS FOUND

- **Furniture is the structural margin weakness.** $5.18M of revenue returns $117.4K of profit — a 2.27% margin — with the damage concentrated in two sub-categories, Tables (–$99.1K) and Bookcases (–$33.6K).

- **Strong sellers can be loss leaders by accident.** The Polycom ViewStation is the #2 product by revenue ($255.3K) and loses $17.8K. Nine other high-revenue products share the pattern.
- **Scale and efficiency are different regions.** South posts the best margin (13.41%) on the smallest revenue base; West posts the worst (8.21%) on the second-largest.
- **Repeat business carries the P&L.** 467 frequent customers — 59% of the base — generate $1.24M, or 82% of total profit.
- **Discounting is not the villain it is usually assumed to be.** Correlation with profit is −0.04; average discount on loss-making lines is 5.2% against 5.0% overall. The apparent "deep discount cliff" rests on two order lines worth $1,002 in total and cannot support a policy conclusion — see §9.
- **Fulfilment is uniform, but not uniformly profitable.** All three ship modes deliver in ~2 days; Delivery Truck carries 42% of revenue at a 4.33% margin against Regular Air's 14.72%.

> MANAGEMENT IMPLICATION
>
> Growth strategy and margin strategy have been substituted for one another. The four-year record shows the business can add revenue without adding profit — 2012 proves it directly. Priority actions are product-level: reprice or retire Tables and Bookcases, put the ten loss-making high-revenue SKUs on a watchlist, and report margin beside revenue in management reporting so the disconnect cannot recur unobserved.

## 02 Business context & objectives

SEVEN QUESTIONS A GENERAL MANAGER WOULD ACTUALLY ASK

Superstore is a US office-products retailer selling furniture, technology and office supplies to corporate, small-business, home-office and consumer buyers across four regions and 48 states. This study turns four years of its transactional record into decision-ready evidence, organised around one governing question: **where is the business creating value, where is it leaking value, and which of those leaks is worth an intervention?**

| THEME | QUESTION PUT TO THE DATA | SECTION |
|---|---|---|
| **Performance** | How much revenue and profit does the business generate, and how stable is that over time? | §5 |
| **Geography** | Which regions, states and cities perform strongly, and which lose money? | §6 |
| **Products** | Which categories, sub-categories and individual products create or destroy profit? | §7 |
| **Customers** | Which segments and accounts generate meaningful revenue — and which of those are unprofitable? | §8 |
| **Pricing** | How is discounting associated with profitability, and where is the real risk? | §9 |
| **Operations** | How consistent is fulfilment, and does it explain any of the margin variance? | §10 |
| **Time** | Is there dependable seasonality, and what is the growth trajectory? | §11 |

INTERPRETATION STANDARD APPLIED THROUGHOUT

Descriptive evidence is kept separate from business interpretation. A high-sales product is not automatically a good product; a low-revenue region is not automatically a weak one; and an observed association is not a cause. Where the evidence is correlational, the report says "associated with" and "warrants investigation" rather than claiming a mechanism.

## 03 Dataset & methodology

FOUR NOTEBOOKS, A NORMALISED DATABASE, AND A DASHBOARD

The project runs as a reproducible pipeline. Each stage writes a validated artefact the next stage consumes, so no cleaning logic is repeated and every downstream figure traces back to a single cleaned file.

01
**Understand**
23 cols · 8,399 rows

02
**Clean & engineer**
→ 33 cols

03
**Explore**
distributions · outliers

04
**Business analysis**
39 questions

05
**SQL layer**
4 tables · 10 queries

06
**Dashboard**
Power BI · 3 pages

| STAGE | TOOLING | PURPOSE | OUTPUT |
|---|---|---|---|
| **Data understanding** | Python · Pandas | Structure, dtypes, missing values, duplicates, ranges, cardinality | `Issue log` |
| **Data cleaning** | Pandas · NumPy | Standardisation, imputation, validation, feature engineering | `33 columns` |
| **EDA** | Matplotlib · Seaborn | Distributions, IQR outliers, correlation, bivariate patterns | `7 findings` |
| **Business analysis** | Pandas | 39 questions across 7 themes, each with evidence and interpretation | `31 charts` |
| **SQL analysis** | PostgreSQL · DBeaver | Normalised schema; window functions, CTEs, percentile segmentation | `10 queries` |
| **Dashboard** | Power BI | Interactive exploration with year / region / category slicers | `3 pages` |

### Grain and scope

One row is a single **order line** — one product within one order — not a whole order. 8,399 lines resolve to 5,496 unique orders placed by 795 customers against a catalogue of 1,263 products. Percentages described as "of orders" in the profitability analysis are therefore percentages of order *lines*; this distinction is preserved wherever it changes the reading.

## 04 Data quality & preparation

PRESERVE BUSINESS MEANING; FIX ONLY WHAT IS GENUINELY BROKEN

The cleaning phase was governed by one rule: an inconvenient number is not a dirty number. No rows were dropped. The 4,264 loss-making lines are the central signal of the study, and removing them would have made the dataset look healthier while destroying the finding.

| ISSUE | EVIDENCE | TREATMENT | RATIONALE |
|---|---|---|---|
| **Missing** `product_base_margin` | 63 (0.75%) | Sub-category median imputation | All 63 trace to 10 SAFCO products with no margin recorded on any row — missing at random against supplier, so a global median would misstate furniture margins. |
| **Duplicate rows** | 0 | None required | Verified before and after cleaning. |
| **Date types** | 2 columns | Cast to `datetime64` | Re-asserted after every CSV round-trip; dtypes do not survive CSV. |
| **Negative profit** | 4,264 (50.77%) | Retained | Real business outcomes, not errors. This is the study's primary signal. |
| **Shipping outlier** | Order #353 — 92 days | Retained, investigated | Isolated logistics anomaly on a $46.94 line; unrelated to price or product. |
| **Margin outlier** | Order #13088 — −1,144% | Retained, investigated | $187.92 sale, −$2,149.80 profit. Shipping cost of $49 cannot explain it; traced to underpricing at $3.25 × 46 units. |
| **Derived fields** | +10 columns | Engineered | Shipping days, profit margin, four date parts, profitability flag, discount and sales bands. |

### The two extreme cases have different causes

Both outliers were traced individually rather than winsorised away, and they turned out to be unrelated failure modes: **#353 is an operations anomaly** (92 days in transit, low priority, trivial value) while **#13088 is a pricing failure** (fulfilment cost roughly eleven times the sale price). That distinction is the first piece of evidence that Superstore's losses have more than one root cause — a theme confirmed independently in §7 and §9.

> QUALITY NOTE CARRIED INTO SQL
>
> Dates and ZIP codes are staged as `TEXT` before conversion, so DD/MM ambiguity and leading-zero loss cannot corrupt the load. The normalised tables reconcile exactly to the cleaned file: 795 customers, 1,263 products, 5,496 orders, 8,399 order items, zero orphaned foreign keys.

## 05 Overall business performance

STABLE REVENUE, DRIFTING PROFIT

| REVENUE | PROFIT | BLENDED MARGIN | AVG ORDER VALUE | BEST YEAR |
| --- | --- | --- | --- | --- |
| **$14.92M** | **$1.52M** | **10.20%** | **$2,714** | **2009** $434,539 |

WEAKEST YEAR

**2012**

$341,902

Across 2009–2012 Superstore recorded **$14,915,600.82** in revenue and **$1,521,767.96** in profit, at a blended margin of **10.20%**. Average order value is $2,713.90 — a high ticket consistent with furniture, technology and bulk supply rather than small-basket retail.

EXHIBIT 1

## Monthly revenue, January 2009 – December 2012

Revenue oscillates in a $200K–$420K band with no sustained trajectory. January 2009 ($516K) is the single highest month and sits well clear of everything after it.




| MONTH | REVENUE |
|---|---|
| Jan 2009 | $516,303 |
| Feb 2009 | $332,481 |
| Mar 2009 | $411,629 |
| Apr 2009 | $393,276 |
| May 2009 | $230,146 |
| Jun 2009 | $263,456 |
| Jul 2009 | $380,504 |
| Aug 2009 | $329,755 |
| Sep 2009 | $325,292 |
| Oct 2009 | $361,555 |
| Nov 2009 | $248,933 |
| Dec 2009 | $415,809 |
| Jan 2010 | $336,527 |
| Feb 2010 | $271,581 |
| Mar 2010 | $217,808 |
| Apr 2010 | $266,969 |
| May 2010 | $283,534 |

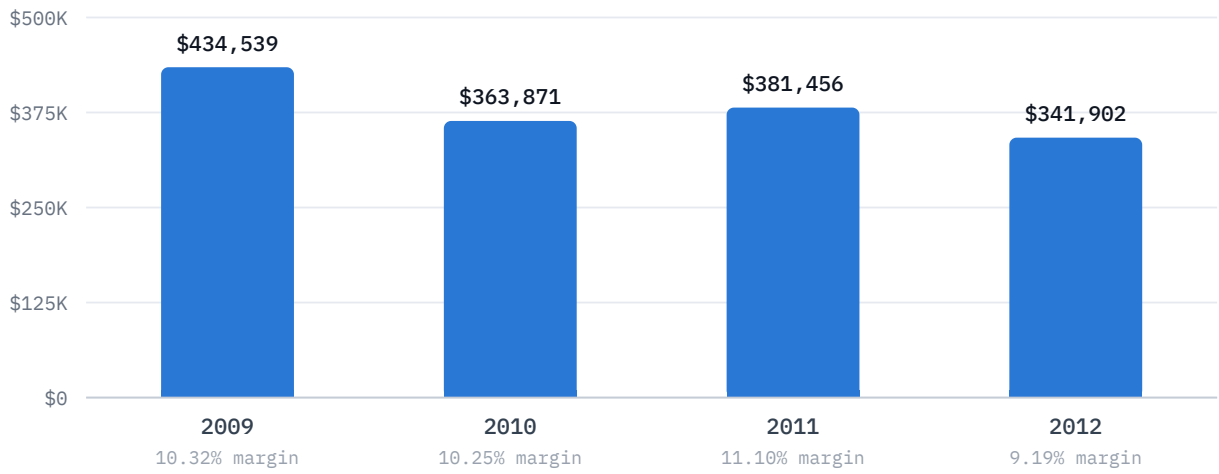
| MONTH | REVENUE |
|---|---|
| Jun 2010 | $293,081 |
| Jul 2010 | $229,886 |
| Aug 2010 | $207,937 |
| Sep 2010 | $418,343 |
| Oct 2010 | $365,252 |
| Nov 2010 | $290,670 |
| Dec 2010 | $368,094 |
| Jan 2011 | $251,467 |
| Feb 2011 | $299,890 |
| Mar 2011 | $296,036 |
| Apr 2011 | $288,213 |
| May 2011 | $262,629 |
| Jun 2011 | $197,741 |
| Jul 2011 | $287,905 |
| Aug 2011 | $274,578 |
| Sep 2011 | $276,050 |
| Oct 2011 | $305,660 |
| Nov 2011 | $367,770 |
| Dec 2011 | $328,877 |
| Jan 2012 | $340,627 |
| Feb 2012 | $276,133 |
| Mar 2012 | $348,208 |
| Apr 2012 | $268,025 |
| May 2012 | $384,588 |
| Jun 2012 | $276,581 |
| Jul 2012 | $242,810 |
| Aug 2012 | $302,745 |
| Sep 2012 | $318,272 |
| Oct 2012 | $351,247 |
| Nov 2012 | $256,020 |
| Dec 2012 | $354,709 |

The series is **volatile rather than trending**. Month-to-month swings dominate any multi-year direction, which points to revenue being driven by the timing of individual large orders rather than by underlying demand growth. On raw monthly totals this is a stable business, not a growing one — an honest reading that §11 tests against quarterly and annual aggregation.

EXHIBIT 2

### Profit and margin by year

Profit fell 21.3% from the 2009 peak to the 2012 trough while revenue held broadly flat — the gap between the two is the study's central theme.




| YEAR | REVENUE | GROWTH | PROFIT | GROWTH | MARGIN | ORDERS |
|---|---|---|---|---|---|---|
| **2009** | $4,209,139 | – | $434,539 | – | 10.32% | 1,394 |
| **2010** | $3,549,681 | −15.67% | $363,871 | −16.26% | 10.25% | 1,393 |
| **2011** | $3,436,817 | −3.18% | $381,456 | +4.83% | 11.10% | 1,317 |
| **2012** | $3,719,964 | +8.24% | $341,902 | −10.37% | 9.19% | 1,392 |
| **Total** | **$14,915,601** | **–** | **$1,521,768** | **–** | **10.20%** | **5,496** |

**2009 has never been matched.** Profit dropped sharply in 2010, recovered partially in 2011 on *falling* revenue — an efficiency gain — then fell again in 2012 on *rising* revenue. Margin moved 11.10% → 9.19% in that final year. Revenue and profit are not merely growing at different rates; in two of three transitions they move in opposite directions.

EVIDENCE

Revenue range across 48 months is $198K–$516K with no trend; profit fell 21.3% from 2009 to 2012 while revenue fell 11.6%.

READING

Demand is stable. Margin is not. The decline is on the cost and pricing side of the business, not the sales side.

IMPLICATION

Volume-led targets will not restore 2009 profit. The remaining sections locate where the margin is going.

## 06 Regional performance

THE LARGEST REGION IS NOT THE MOST EFFICIENT ONE

| REGIONS | LARGEST | BEST MARGIN | WORST MARGIN | MARGIN SPREAD |
|---|---|---|---|---|
| 4 | Central<br>$4.70M revenue | 13.41%<br>South | 8.21%<br>West | 5.20pp |

LOSS-MAKING STATES

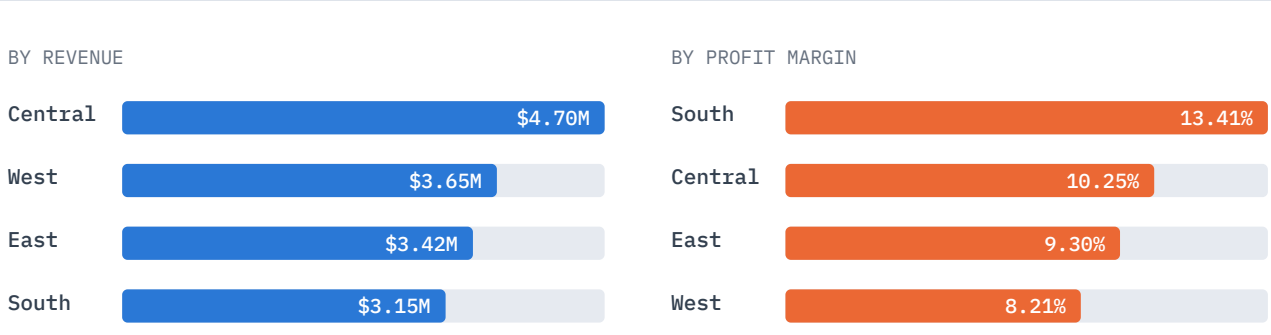
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Montana, −$9,127

EXHIBIT 3

### Revenue and margin by region — the ranking inverts

Two views of the same four regions, ordered independently. Central leads on scale; South leads on efficiency; West appears high on the left and last on the right.



| REGION | REVENUE | PROFIT | MARGIN | LINES | REV. RANK | MARGIN RANK |
|---|---|---|---|---|---|---|
| **Central** | $4,699,167 | $481,891 | 10.25% | 2,710 | 1 | 2 |
| **West** | $3,649,748 | $299,518 | 8.21% | 1,956 | 2 | 4 |
| **East** | $3,416,466 | $317,852 | 9.30% | 1,895 | 3 | 3 |
| **South** | $3,150,219 | $422,507 | 13.41% | 1,838 | 4 | 1 |

South converts sales into profit **63% more efficiently than West** (13.41% against 8.21%) on 14% less revenue. Central's dollar leadership comes from volume, not efficiency — it sits mid-table on margin. Ranking regions by revenue, as most sales reporting does, would place West second and conceal the largest regional margin gap in the business.

### State and city detail

**Illinois** is the most profitable state ($108,532), ahead of Texas, Ohio, California and New York in a tight $86K–$94K band. **Montana is the only state with a net loss** — −$9,127 on $76,072 of sales, a −12.00% margin. That is not a small-sample artefact: $76K is real volume, and every dollar sold there cost the business money on average. **Madison** ($119,943) and **San Francisco** ($103,079) are the only cities above $100K, after which the ranking drops to a $53K–$66K band.

> WHY THIS MATTERS OPERATIONALLY
>
> Because the loss is confined to a single state rather than spread across a region, the appropriate response is a targeted order- and product-mix review in Montana — not a regional restructuring. Conversely, West's margin gap *is* regional in scale and warrants investigation of its product mix and pricing, since §10 rules out shipping speed as the cause.

EVIDENCE

Revenue rank and margin rank disagree for three of four regions. South is 4th on revenue and 1st on margin; West is 2nd and 4th.

READING

Regional revenue reflects market size; regional margin reflects what is sold there and at what price. They are independent measures.

IMPLICATION

Closing West's gap to South's margin on its existing revenue base would be worth roughly **$190K** a year — larger than the entire Montana loss.

## 07 Product & category performance

WHERE THE MONEY IS ACTUALLY LOST

| CATEGORIES | SUB-CATEGORIES | BEST MARGIN | WEAKEST MARGIN | LOSS-MAKING SUB-CATS |
|---|---|---|---|---|
| 3 | 17 | 14.81% Technology | 2.27% Furniture | 4 −$140,547 |

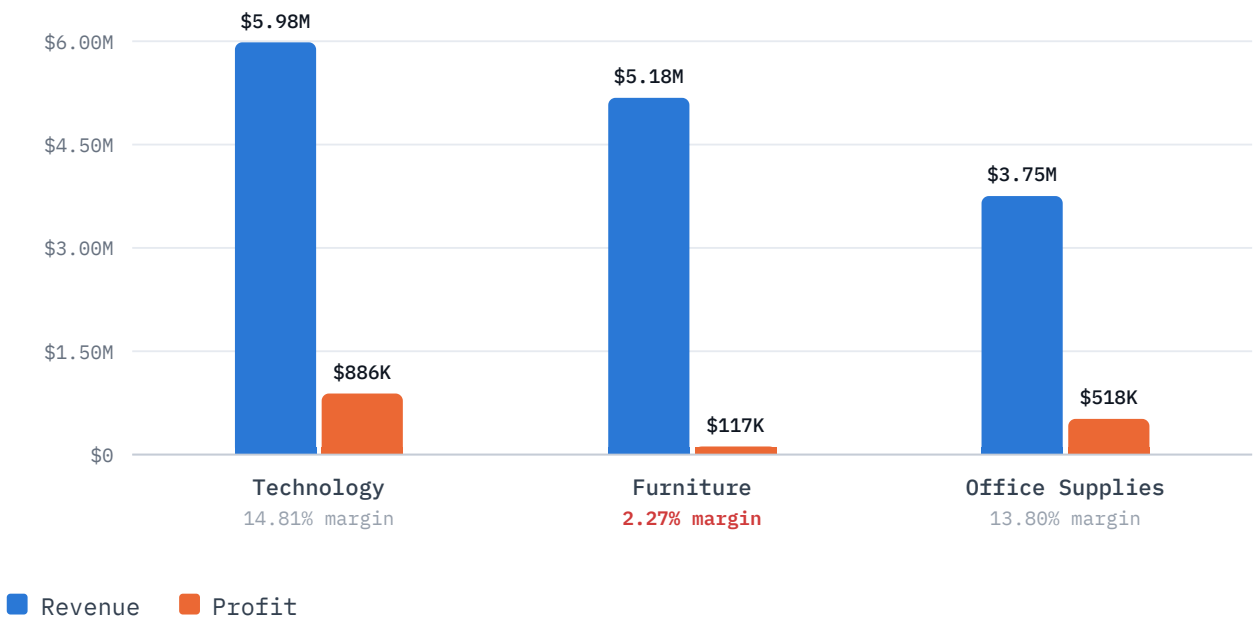
| LOSS-MAKING TOP SKUS |
|---|
| 10 −$141,149 |

Product analysis produces the sharpest evidence in the study. Technology and Furniture sell at comparable scale. They do not return comparable profit.

EXHIBIT 4

### Revenue against profit by category

Both series are US dollars on one shared scale. Furniture's revenue bar is 85% of Technology's; its profit bar is 13% of it.



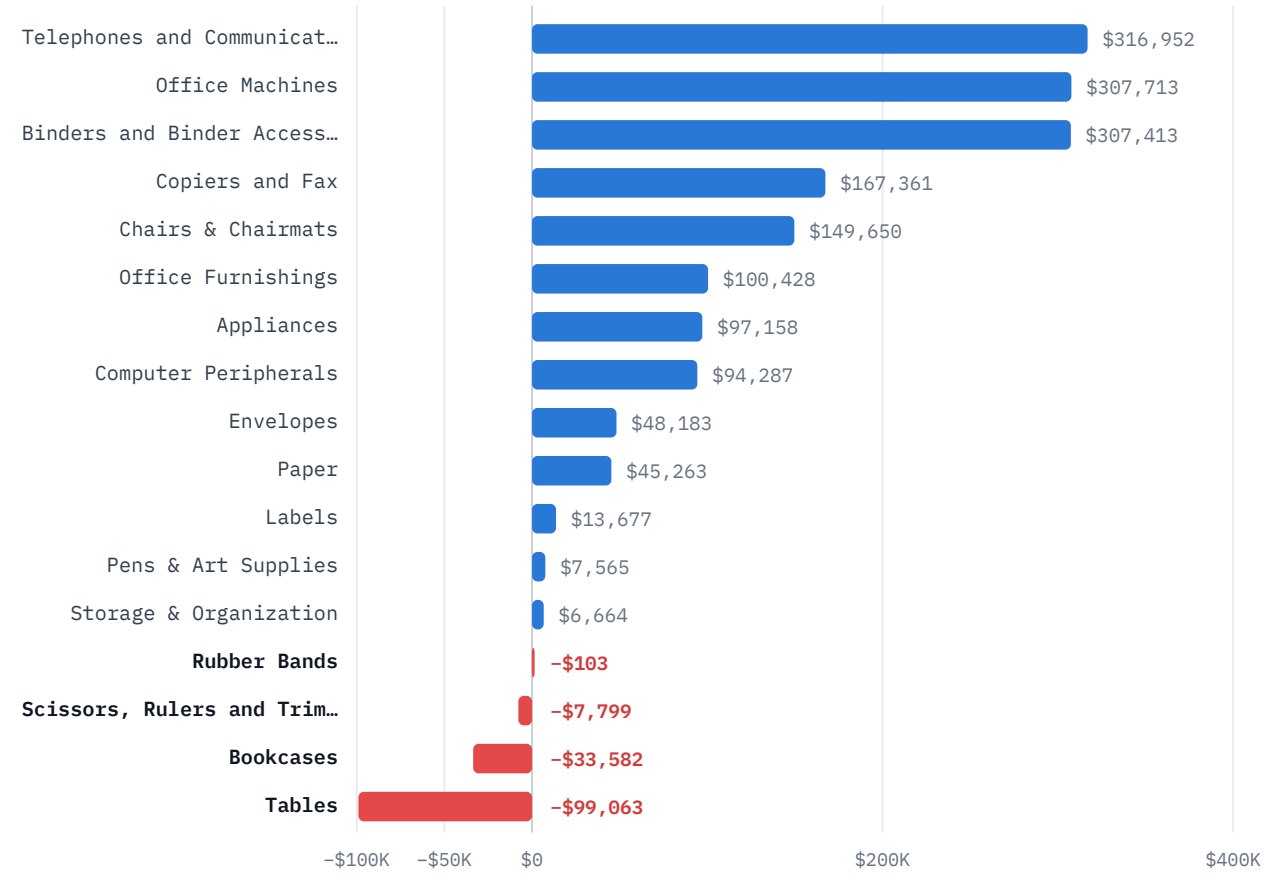
| CATEGORY | REVENUE | PROFIT | MARGIN | LINES | PROFIT PER LINE |
|---|---|---|---|---|---|
| **Technology** | $5,984,248 | $886,314 | 14.81% | 2,065 | $429 |
| **Office Supplies** | $3,752,762 | $518,021 | 13.80% | 4,610 | $112 |
| **Furniture** | $5,178,591 | $117,433 | 2.27% | 1,724 | $68 |

Exploratory analysis had flagged Office Supplies as the priority problem — it carries 54.9% of all order lines and the single worst individual loss in the dataset. Category-level margin revises that reading. **Office Supplies is a high-volume, distributed problem at a respectable 13.80% margin; Furniture is a concentrated, structural one at 2.27%.** Both matter, for different reasons and with different remedies.

EXHIBIT 5

## Profit by sub-category — four destroy value

All 17 sub-categories, ordered by total profit. Red bars fall below zero. Tables and Bookcases are both Furniture, and together account for 94% of all sub-category losses.




| SUB-CATEGORY | TOTAL PROFIT | MARGIN |
|---|---|---|
| **Telephones and Communication** | $316,952 | 16.78% |
| **Office Machines** | $307,713 | 14.19% |
| **Binders and Binder Accessories** | $307,413 | 30.05% |
| **Copiers and Fax** | $167,361 | 14.81% |
| **Chairs & Chairmats** | $149,650 | 8.49% |
| **Office Furnishings** | $100,428 | 14.39% |
| **Appliances** | $97,158 | 13.18% |
| **Computer Peripherals** | $94,287 | 11.85% |
| **Envelopes** | $48,183 | 27.68% |
| **Paper** | $45,263 | 10.14% |

| SUB-CATEGORY | TOTAL PROFIT | MARGIN |
|---|---|---|
| **Labels** | $13,677 | 35.09% |
| **Pens & Art Supplies** | $7,565 | 4.53% |
| **Storage & Organization** | $6,664 | 0.62% |
| **Rubber Bands** | −$103 | -0.68% |
| **Scissors, Rulers and Trimmers** | −$7,799 | -9.63% |
| **Bookcases** | −$33,582 | -4.08% |
| **Tables** | −$99,063 | -5.22% |

The four loss-makers behave differently and deserve different responses:

- **Tables** — −$99,063 across 361 lines at a −5.22% margin. The largest loss on every measure, spread over hundreds of separate transactions. This is structural, not anomalous.
- **Bookcases** — −$33,582 across 189 lines at −4.08%. Same pattern, smaller scale.
- **Scissors, Rulers and Trimmers** — the *worst rate* at −9.63%, though only −$7,799 in absolute terms. Proportionally the least viable line in the catalogue.
- **Rubber Bands** — −$103 at −0.68%. Effectively break-even; not a priority.

### High-revenue products that lose money

The pattern also holds at SKU level, and this is where it becomes actionable. Ten products with above-median revenue carry negative total profit:

| PRODUCT | REVENUE | PROFIT | IMPLIED MARGIN |
|---|---|---|---|
| **Polycom ViewStation ISDN Videoconferencing Unit** | $255,304 | −$17,771 | −6.96% |
| **Bretford CR8500 Meeting Room Furniture** | $147,412 | −$1,990 | −1.35% |
| **Canon imageCLASS 2200 Advanced Copier** | $114,173 | −$31,886 | −27.93% |
| **Chromcraft Bull-Nose Wood Rectangular Table** | $106,629 | −$6,345 | −5.95% |
| **BoxOffice By Design Meeting Tables** | $104,483 | −$14,939 | −14.30% |
| **Hon 94000 Series Round Tables** | $102,006 | −$4,497 | −4.41% |
| **Bretford Rectangular Conference Table Tops** | $91,730 | −$2,795 | −3.05% |
| **Polycom ViewStation H323 Adapter** | $88,273 | −$1,007 | −1.14% |
| **Epson DFX-8500 Dot Matrix Printer** | $85,975 | −$29,400 | −34.20% |
| **Global High-Back Leather Tilter, Burgundy** | $82,752 | −$30,519 | −36.88% |

Two observations make this list the highest-value output of the study. First, the **Polycom ViewStation is the #2 best-selling product in the entire catalogue** — the loss is invisible in any revenue-ranked report. Second, the list spans both Furniture *and* Technology, the strongest category by margin, so a category-level remedy would miss half of it. These are pricing decisions, not merchandising ones: real demand exists at the current price, and the current price is below cost.

HIGHEST-PRIORITY ACTION

Three SKUs — the Canon imageCLASS 2200, the Epson DFX-8500 and the Global High-Back Leather Tilter — lose more than $270 per $1,000 of revenue sold. Combined, these ten products destroy **$141,149** of profit, equal to 9.3% of the four-year total. A price review on ten line items is a materially smaller intervention than any of the strategic options elsewhere in this report.

EVIDENCE

Four sub-categories lose **$140,547**; ten high-revenue SKUs lose **$141,149**. Furniture returns 2.27% on revenue 85% the size of Technology's.

READING

The losses are concentrated, identifiable and repeat across hundreds of transactions — structural pricing, not isolated bad deals.

IMPLICATION

The two lists overlap — five of the ten SKUs sit inside Tables — so their union, not their sum, is the exposure: **$251,130**, or **16.5%** of four-year profit, reachable through a handful of product decisions.

## 08 Customer performance

RETENTION ALREADY CARRIES THE P&L

| CUSTOMERS | SEGMENTS | ORDERS PER CUSTOMER | REVENUE PER CUSTOMER | PROFIT FROM REPEAT |
|---|---|---|---|---|
| 795 | 4 | 6.9<br>4-year average | $18,762<br>lifetime average | 81.6%<br>467 customers |

TOP ACCOUNT

**$34,005**

Emily Phan

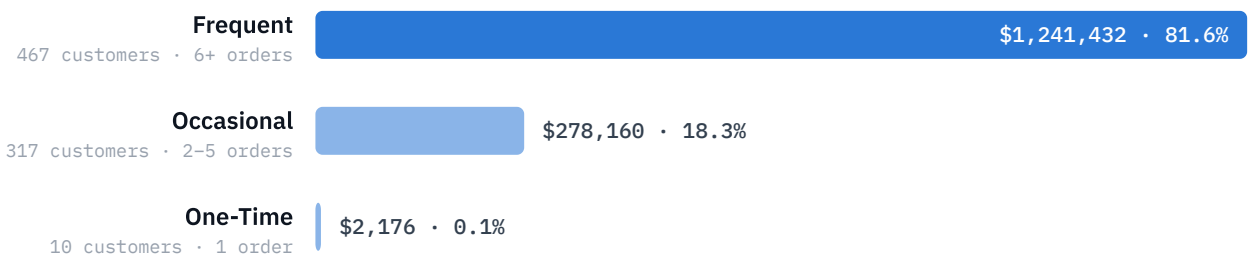
| SEGMENT | REVENUE | PROFIT | MARGIN | LINES | READING |
|---|---|---|---|---|---|
| **Corporate** | $5,498,905 | $599,746 | 10.91% | 3,076 | Largest on both revenue and profit — the scale account |
| **Home Office** | $3,564,764 | $318,354 | 8.93% | 2,032 | Second on revenue, weakest on margin |
| **Consumer** | $3,063,611 | $287,960 | 9.40% | 1,649 | Mid-scale, below-average efficiency |
| **Small Business** | $2,788,321 | $315,708 | 11.32% | 1,642 | Smallest revenue, **best margin** |

The regional inversion repeats exactly at segment level: **Small Business is the least valuable segment by revenue and the most efficient by margin** (11.32%), while Home Office is second-largest and least efficient (8.93%). Corporate should remain the priority for scale — it produces 39% of all profit — but Small Business's economics are the ones worth understanding and, if possible, replicating at volume.

EXHIBIT 6

### Profit contribution by purchase frequency

Customers classified by distinct order count in the SQL layer: One-Time (1), Occasional (2–5), Frequent (6+). Frequent buyers are 59% of the base and 82% of profit.




| COHORT | CUSTOMERS | SHARE OF BASE | TOTAL PROFIT | SHARE OF PROFIT | AVG PER CUSTOMER |
|---|---|---|---|---|---|
| **Frequent (6+ orders)** | 467 | 58.7% | $1,241,432 | 81.6% | $2,658 |
| **Occasional (2–5)** | 317 | 39.9% | $278,160 | 18.3% | $877 |
| **One-Time (1)** | 10 | 1.3% | $2,176 | 0.1% | $218 |

A frequent customer is worth **3.0× an occasional one and 12.2× a one-time buyer** in lifetime profit. Acquisition is not where this business's profit comes from; repeat relationships are. That makes the

next finding a genuine risk rather than a curiosity.

### High-revenue accounts running at a loss

**Emily Phan** ($117,124 revenue, $34,005 profit) and **Deborah Brumfield** ($97,433 / $31,121) lead both the revenue and profit rankings — unambiguously the two accounts most worth protecting. Below them the two lists diverge sharply. Roy Skaria and Sylvia Foulston are top-five by revenue and absent from the top ten by profit, while Karen Carlisle, John Stevenson and Clytie Kelty rank highly on profit without appearing in the revenue top ten.

The SQL layer isolates the risk directly: customers in the **top 20% by revenue whose average line margin is below the company average**. Every flagged account carries a negative average margin.

| CUSTOMER | TOTAL REVENUE | AVG LINE MARGIN | NOTE |
|---|---|---|---|
| **Deborah Brumfield** | $97,433 | −29.77% | Also #2 by total profit — profitable in aggregate, but with a loss-making tail |
| **Roy Skaria** | $92,542 | −18.71% | Top-5 revenue, outside top-10 profit |
| **Darren Budd** | $84,606 | −21.02% | — |
| **Sylvia Foulston** | $83,277 | −19.97% | Top-5 revenue, outside top-10 profit |
| **Julia West** | $44,073 | −20.62% | **Net loss of −$14,154** — the largest customer-level loss |
| **Roy Phan** | $46,546 | −31.68% | Net loss of −$9,614 |
| **Jay Fine** | — | −82.77% | Worst average margin among top-20% spenders |

> READ THIS CAREFULLY
>
> Average *line* margin and total profit are different measures and can disagree — Deborah Brumfield appears on both the "most profitable" and "negative average margin" lists without contradiction, because a few large profitable lines outweigh many small loss-making ones. The flag identifies accounts whose **typical transaction** loses money, which is the right target for a terms review; it is not a list of customers to drop.

EVIDENCE

467 frequent customers produce $1.24M of $1.52M total profit. Separately, every top-quintile spender tested against the company benchmark shows a negative average line margin.

READING

Profit is concentrated in repeat relationships, and revenue concentration does not track it — the two lists overlap only at the very top.

IMPLICATION

Retention protects 82% of profit and currently has no dedicated mechanism. Account reviews should be triggered by margin, not by spend.

## 09 Discounting & profitability

THE FINDING THAT SURVIVED, AND THE ONE THAT DID NOT

| AVERAGE DISCOUNT | MAXIMUM DISCOUNT | CORRELATION W/ PROFIT | LINES IN LOW BAND | ZERO-DISCOUNT MARGIN |
|---|---|---|---|---|
| 4.97% | 25.0% | −0.037 negligible | 82.07% 6,893 lines | 12.60% |

MEDIUM-BAND MARGIN

6.50%

−6.10pp

The intuitive explanation for a 50.77% loss rate is discount abuse. The data does not support it, and testing that hypothesis properly is one of the more important things this study does.

CORRELATION

Discount against profit: **−0.037**. Against margin: **−0.058**. A genuine driver would sit near −0.5.

LOSS-MAKING VS PROFITABLE

Average discount on loss-making lines **5.19%** against **4.74%** on profitable ones — a gap of 0.45pp.

ACROSS SUB-CATEGORIES

Average discount ranges **4.80%–5.15%** across all 17 sub-categories — a spread of 0.35pp.

Discounting is applied almost uniformly across the product range, and profitability outcomes vary enormously *within* that uniform treatment. Since the input barely varies and the output varies hugely, the input cannot be the explanation. **Discount policy is exonerated as a broad driver of**

**losses.** The evidence points instead to product-level cost and pricing structure — exactly what §7 and the Order #13088 investigation found independently.

EXHIBIT 7

### Margin by discount band — and why the last bar cannot be trusted

Bars show revenue-weighted margin (profit ÷ sales) per band. Bar width is scaled by band revenue, so the visual weight reflects how much business each band represents.




| BAND | LINES | REVENUE | PROFIT | WEIGHTED MARGIN | WEIGHT IN BUSINESS |
|---|---|---|---|---|---|
| **No discount (0%)** | 756 | $1,493,748 | $188,189 | 12.60% | 10.0% of revenue |
| **Low (0.1–9.9%)** | 6,893 | $12,274,892 | $1,259,617 | 10.26% | 82.3% of revenue |
| **Medium (10–19.9%)** | 748 | $1,145,958 | $74,461 | 6.50% | 7.7% of revenue |
| **High (20%+)** | 2 | $1,002 | −$499 | −49.76% | 0.007% OF REVENUE |

CORRECTION TO A TEMPTING HEADLINE

The −49.76% margin in the High band is arithmetically correct and analytically meaningless. It is computed from **two order lines totalling $1,002.30 in revenue** — 0.007% of the business. A "deep discounting destroys margin" conclusion drawn from two transactions would not survive review, and this report does not draw it.

The defensible discount finding is the **Medium band**: 748 lines and $1.15M of real revenue, where weighted margin falls to 6.50% against 12.60% at zero discount. That is a **6.1-point degradation across a meaningful slice of the business**, and it is the finding that should drive policy.

The maximum discount anywhere in the dataset is **25%**, and 82.07% of all lines sit in the Low band. Superstore is not a heavily discounting business. Among the ten deepest-discount orders, six are unprofitable and four are not — including a calculator discounted 10% that returned $2,052 on $5,196. Discount level alone does not determine the outcome; the product's underlying economics do.

> PRICING RECOMMENDATION
>
> Do not introduce a blanket discount ceiling — the data gives no support for one. Introduce a **conditional control**: require approval where a discount above 10% is applied to a product whose trailing margin is already below the category average. That targets the Medium-band degradation and the §7 SKU list simultaneously, without restricting the 82% of business that is priced sensibly today.

## 10 Shipping & operations

CONSISTENT DELIVERY, INCONSISTENT ECONOMICS

| AVG SHIPPING TIME | MODE SPREAD | REGION SPREAD | DELAYED LINES | BEST MODE MARGIN |
|---|---|---|---|---|
| 2.03 days | 0.04 days<br>fastest to slowest | 0.14 days<br>1.95 – 2.09 | 105<br>1.25%, >8.94 days | 14.72%<br>Regular Air |

WORST MODE MARGIN

4.33%

Delivery Truck

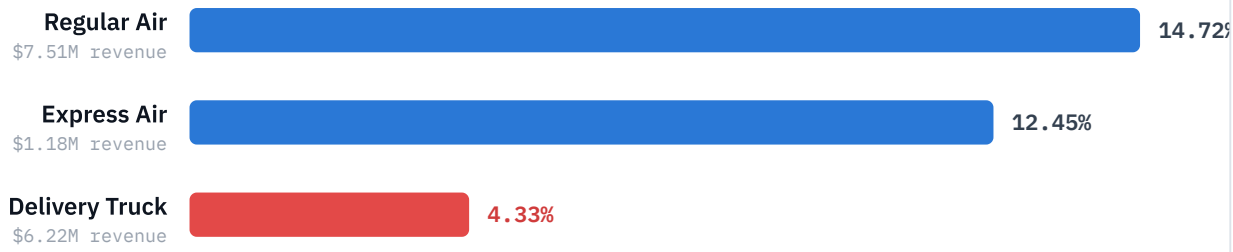
Operationally, Superstore is remarkably uniform. **All three shipping modes average almost exactly two days** — Express Air 2.00, Regular Air 2.04, Delivery Truck 2.04 — and all four regions fall between 1.95 and 2.09 days. Shipping time has near-zero correlation with every other numeric variable in the dataset.

Two consequences follow. First, **shipping speed cannot explain the regional margin gap**: South is fastest (1.95 days) and West slowest (2.09), a 0.14-day difference far too small to account for a 5.2-point margin spread. Second, and more awkwardly for the business, **"Express Air" is not measurably faster than "Regular Air"** despite a name and presumably a cost that imply otherwise.

EXHIBIT 8

### Margin by shipping mode — a gap the volume ranking hides

Regular Air's profit leadership is a scale effect (75% of lines). Margin tells a different story: Delivery Truck moves 42% of all revenue at less than a third of Regular Air's efficiency.




| MODE | LINES | REVENUE | PROFIT | MARGIN | AVG DAYS |
|---|---|---|---|---|---|
| **Regular Air** | 6,270 | $7,506,303 | $1,104,693 | 14.72% | 2.04 |
| **Delivery Truck** | 1,146 | $6,224,879 | $269,645 | 4.33% | 2.04 |
| **Express Air** | 983 | $1,184,419 | $147,430 | 12.45% | 2.45 |

AN EXTENSION BEYOND THE NOTEBOOK ANALYSIS

The notebooks compared shipping modes on *total profit*, where Regular Air's 75% volume share makes it the obvious leader. Recomputing on margin — from the same cleaned dataset — surfaces a finding the volume view conceals: **Delivery Truck carries $6.22M of revenue, 42% of the business, at a 4.33% margin.** Delivery Truck is the mode used for bulky Furniture, so this is very likely the same Tables-and-Bookcases problem observed through a different lens rather than an independent one — but it gives operations a concrete handle on it, and the overlap is worth quantifying directly.

## The delay tail

Using a threshold of 8.94 days (three standard deviations above the mean), **105 lines — 1.25% of the dataset** — qualify as unusually delayed. Order #353 (92 days) and Order #1699 (84 days) are the extremes; after them the distribution drops sharply to 31, 28 and 27 days, with the bulk clustered near the threshold itself. Regular Air appears most often among the delays, but it also handles 75% of shipments, so this is a volume artefact rather than evidence of unreliability.

The SQL benchmark query adds one operational lead worth following: while Furniture averages ~2 days like everything else, a cluster of Furniture orders consistently ships in 7–9 days, with a worst case at 17 days. That is a specific fulfilment pattern, not a company-wide problem.

EVIDENCE

Delivery times vary by 0.14 days across regions and 0.04 across modes; `shipping_days` correlates ≈0.00 with every financial variable.

READING

Fulfilment speed is effectively constant and can be excluded as an explanation for the margin variance found in §6 and §7.

IMPLICATION

Ruling a cause out is a result. It concentrates the remaining investigation on price and product mix — and raises whether Express Air's premium buys anything measurable.

## 11 Time trends & seasonality

A SEASONAL PATTERN THAT DOES NOT REPEAT

| STRONGEST QUARTER | WEAKEST QUARTER | STRONGEST MONTH | WEAKEST MONTH | 2012 REVENUE |
|---|---|---|---|---|
| Q4 | Q2 | December | June | +8.24% |
| $4.01M · 10.58% | $3.41M | $1.47M pooled | $1.03M pooled | |

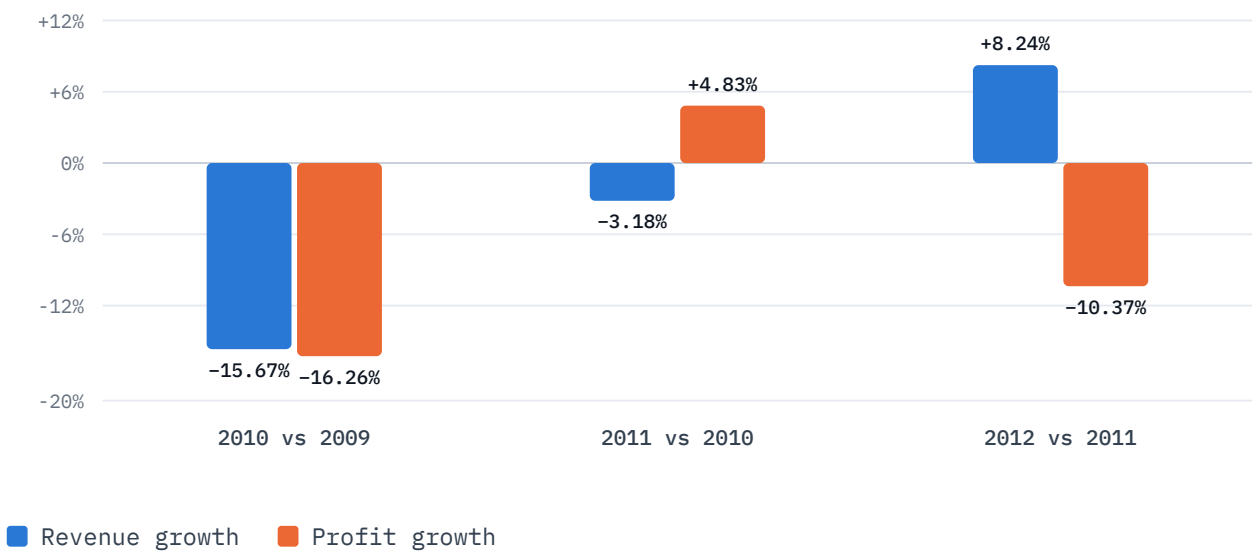
2012 PROFIT

−10.37%

EXHIBIT 9

### Year-over-year growth: revenue against profit

In two of three year transitions the two measures move in opposite directions. 2012 is the clearest case — and the most recent.

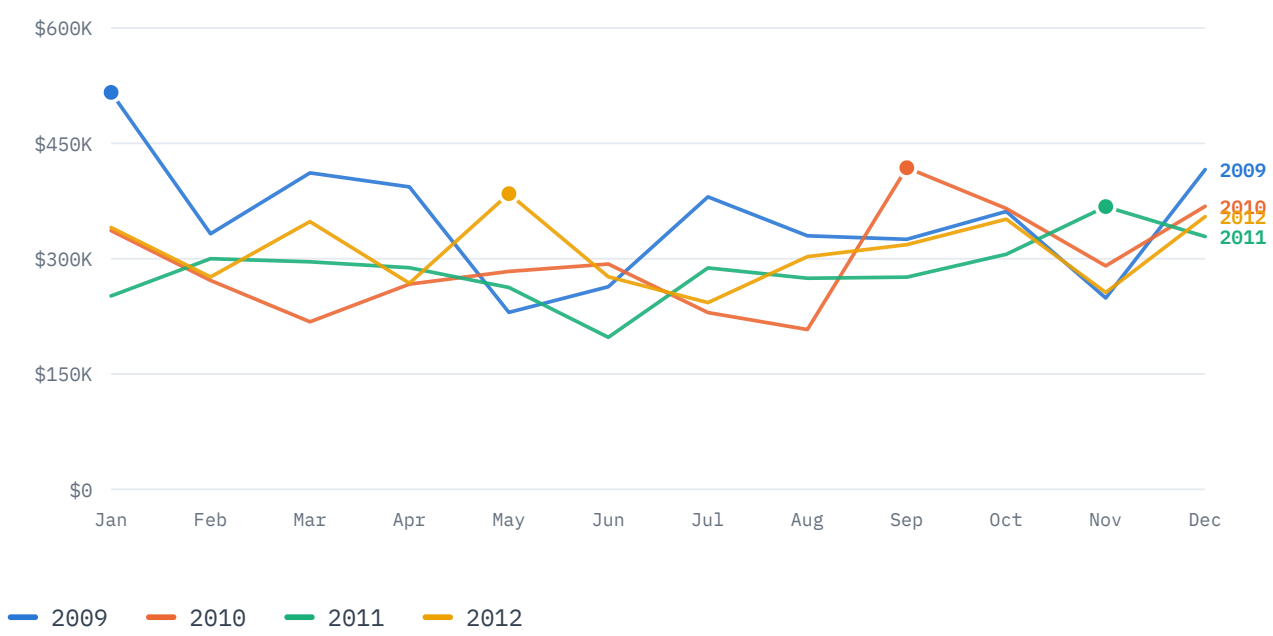



**2012 is the management warning in this dataset.** Revenue grew 8.24% and order volume rose from 1,317 to 1,392, yet profit fell 10.37% and margin dropped from 11.10% to 9.19%. More business was written, and less money was made. Everything else in this report is an explanation of how that happens.

EXHIBIT 10

### Monthly sales by year — the peak moves every year

2009 peaks in January, 2010 in September, 2011 in November, 2012 in May. Four years, four different peak months.




| MONTH | 2009 | 2010 | 2011 | 2012 |
|---|---|---|---|---|
| **Jan** | $516,303 | $336,527 | $251,467 | $340,627 |
| **Feb** | $332,481 | $271,581 | $299,890 | $276,133 |
| **Mar** | $411,629 | $217,808 | $296,036 | $348,208 |
| **Apr** | $393,276 | $266,969 | $288,213 | $268,025 |
| **May** | $230,146 | $283,534 | $262,629 | $384,588 |
| **Jun** | $263,456 | $293,081 | $197,741 | $276,581 |
| **Jul** | $380,504 | $229,886 | $287,905 | $242,810 |
| **Aug** | $329,755 | $207,937 | $274,578 | $302,745 |
| **Sep** | $325,292 | $418,343 | $276,050 | $318,272 |
| **Oct** | $361,555 | $365,252 | $305,660 | $351,247 |
| **Nov** | $248,933 | $290,670 | $367,770 | $256,020 |
| **Dec** | $415,809 | $368,094 | $328,877 | $354,709 |

Aggregated across all four years, December is the strongest month ($1.47M) and June the weakest ($1.03M) — a 30% spread — and Q4 leads on both revenue ($4.01M) and profit ($424,876) while Q2

trails on both. That is a real pattern in the pooled data.

It is also **not a reliable planning input**. No single month peaks consistently, and January's high aggregate rank is carried almost entirely by an exceptional January 2009. The honest conclusion is that Superstore exhibits month-to-month volatility rather than a dependable seasonal calendar. Stating "always build inventory before December" would over-read four years of noisy data; the pattern should be re-validated on a longer series before it drives commitments.

| QUARTER | REVENUE | PROFIT | MARGIN | READING |
|---|---|---|---|---|
| **Q1** | $3,898,688 | $378,749 | 9.71% | Second on revenue, third on profit |
| **Q2** | $3,408,238 | $358,239 | 10.51% | Weakest quarter on both measures |
| **Q3** | $3,594,077 | $359,903 | 10.01% | $304K less revenue than Q1, near-identical profit |
| **Q4** | $4,014,598 | $424,876 | 10.58% | Strongest on revenue, profit and margin |

Even here the disconnect appears: Q1 out-sells Q3 by $304,611 and out-earns it by $18,846. Revenue is a poor predictor of profit at every level of aggregation tested in this study.

EVIDENCE

Four years produce four different peak months. 2012 grew revenue 8.24% and lost 10.37% of profit, with margin falling 11.10% → 9.19%.

READING

The pooled seasonal pattern is an artefact of averaging, not a repeating cycle. The growth record is volatile in both directions.

IMPLICATION

Do not plan inventory on the pooled month ranking. Do treat 2012 as evidence that revenue targets alone are an insufficient control.

## 12 The SQL analysis layer

POSTGRESQL · NORMALISED SCHEMA · 10 ADVANCED QUERIES

The cleaned file was loaded into PostgreSQL and normalised from a flat 33-column staging table into four related tables. Staging keeps dates and ZIP codes as `TEXT` so that DD/MM ambiguity and leading-zero truncation cannot corrupt the import; conversion to `DATE` happens on the way into the normalised layer.

| customers | 795 |
| --- | --- |
| **customer_id · PK** | |
| customer_name | |
| city · state · zip_code | |
| region | |
| customer_segment | |

| products | 1,263 |
| --- | --- |
| **product_id · PK** | |
| product_name | |
| product_category | |
| product_sub_category | |
| product_container | |
| product_base_margin | |

| orders | 5,496 |
| --- | --- |
| **order_id · PK** | |
| customer_id → customers | |
| order_date · ship_date | |
| shipping_days | |
| order_priority · ship_mode | |
| year · month · quarter · dow | |

| order_items | 8,399 |
| --- | --- |
| **row_id · PK** | |
| order_id → orders | |
| product_id → products | |
| order_quantity · unit_price | |
| sales · discount · profit | |
| profit_margin · is_profitable | |
| discount_band · sales_band | |

Referential integrity was verified after population: **zero orders with a null customer link and zero order items with a null product link**, and all four row counts reconcile to the source file.

### Representative query — high-value, high-risk customers

This is the query behind the customer risk table in §8. It combines three CTEs, a percentile threshold and a company-wide benchmark to isolate top-quintile spenders whose typical transaction loses money.

```sql
-- Top-20% spenders whose average line margin trails the company average
with customer_metrics as (
    select c.customer_id, c.customer_name,
           sum(oi.sales) as total_sales,
           avg(oi.profit_margin) as avg_margin
    from order_items oi
    join orders o    on o.order_id = oi.order_id
    join customers c on c.customer_id = o.customer_id
    group by c.customer_id, c.customer_name
),
sales_threshold as (
    select percentile_cont(0.8) within group (order by total_sales) as p80_sales
    from customer_metrics
),
company_avg as (
    select avg(profit_margin) as avg_margin from order_items
)
select cm.customer_name,
       round(cm.total_sales, 2) as total_sales,
       round(cm.avg_margin, 2)  as avg_margin
from customer_metrics cm, sales_threshold st, company_avg ca
where cm.total_sales >= st.p80_sales
  and cm.avg_margin  < ca.avg_margin
order by cm.total_sales desc;
```

### Representative query — top 3 products within each category

A ranked CTE, used because window functions cannot be filtered in `WHERE`:

```sql
with ranked_products as (
    select p.product_category, p.product_name,
           sum(oi.profit) as total_profit,
           row_number() over (
               partition by p.product_category
               order by sum(oi.profit) desc
           ) as profit_rank
    from order_items oi
    join products p on oi.product_id = p.product_id
    group by p.product_category, p.product_name
)
select product_category, product_name, total_profit, profit_rank
from ranked_products
where profit_rank <= 3
order by product_category, profit_rank;
```

## All ten queries

| # | QUESTION | TECHNIQUE DEMONSTRATED | FINDING |
|---|---|---|---|
| 1 | **Customer lifetime value ranking** | `RANK()` vs `DENSE_RANK()` side by side | Emily Phan leads at $34,005 — nearly double the #5 account |
| 2 | **Running total & 3-month moving average** | Window frame `ROWS BETWEEN 2 PRECEDING` | Cumulative revenue reaches $14.92M; the smoothed trend stays flat at $300–330K |
| 3 | **Top 3 products per category** | Ranked CTE + `PARTITION BY` | Profit within a category concentrates in a few items — Furniture's #1 out-earns #2 and #3 combined |
| 4 | **MoM and YoY revenue growth** | `LAG()` at offsets 1 and 12, divide-by-zero guarded | YoY swings from −30% to +46% within 2012 alone |
| 5 | **High-value, high-risk customers** | Multi-CTE + `percentile_cont(0.8)` | Every top-quintile spender flagged carries a negative average margin |
| 6 | **Sub-category rank within region** | Dual `RANK()` windows, asc and desc | Labels is best in all four regions (~26–28%); Appliances worst in three |
| 7 | **Discount break-even threshold** | `GROUP BY` + `HAVING` on an aggregate | Average line margin is negative in every band — including zero discount |
| 8 | **Order frequency & retention cohorts** | Chained CTEs + `CASE` segmentation | 467 frequent customers generate $1.24M — 82% of profit |
| 9 | **Shipping vs category benchmark** | Derived-table join + ranked deviation | Furniture averages ~2 days but a cluster ships in 7–9, worst case 17 |
| 10 | **Executive KPI snapshot, YoY** | Conditional aggregation, single-pass | 2012 revenue +8.2% and orders up, profit down; margin 11.10% → 9.19% |

A NOTE ON QUERY 6 — THE STRONGEST INDEPENDENT RESULT

**Labels is the best-margin sub-category in all four regions** (25.95%–27.76%) and **Appliances the worst in three of four**, reaching −59.64% in the East. Consistency across regions is what makes this useful: a problem that appears everywhere is a pricing or cost problem with the product, not a regional execution problem — and it points to two concrete candidates, one to expand and one to reprice.

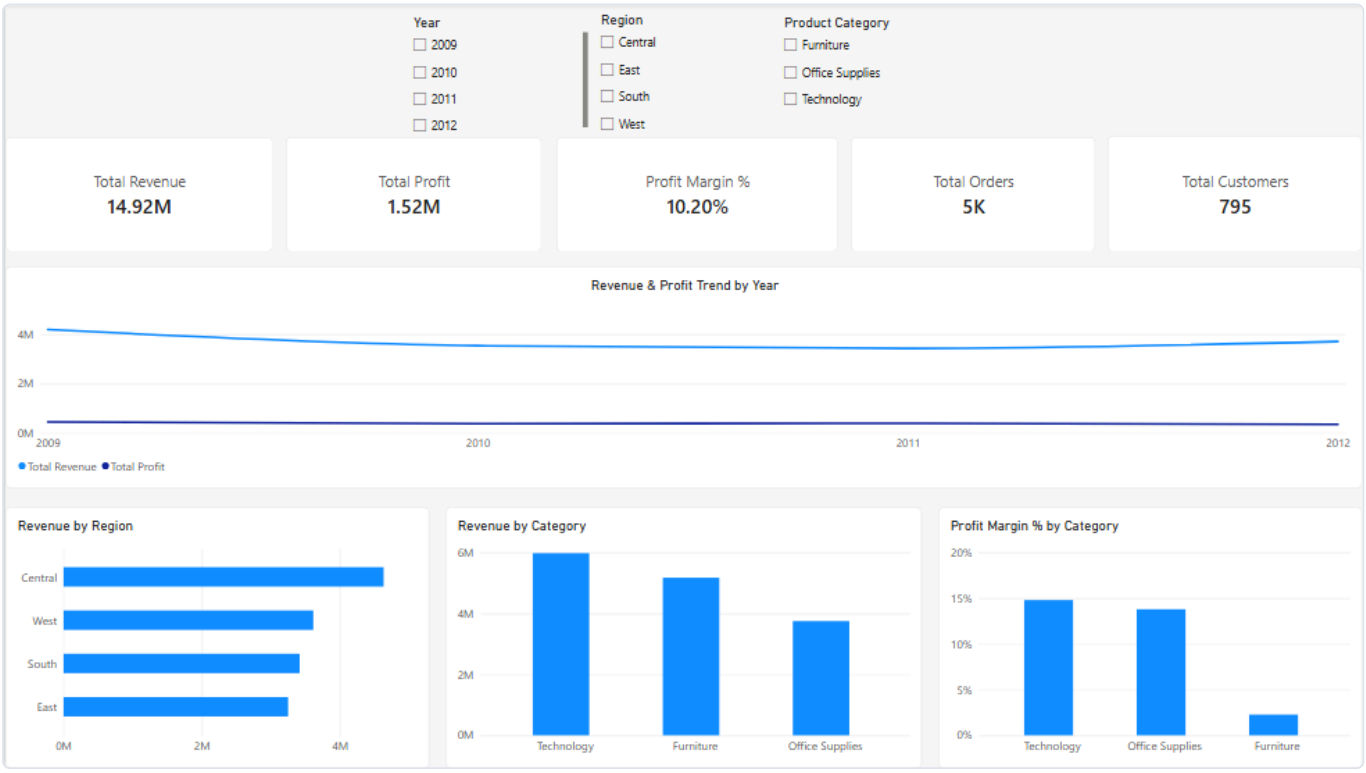
TWO MEASURES OF MARGIN, DELIBERATELY DISTINGUISHED

Query 7 reports the *average of line-level margins* and finds every discount band negative, including zero-discount at −7.4%. §9 reports *revenue-weighted* margin and finds every band except High positive. Both are correct; they answer different questions. The simple average treats a $6 line and a $60,000 line equally, so it describes the **typical transaction**; the weighted figure describes the **business**. This report uses weighted margin for financial statements and flags the unweighted figure wherever it is the one quoted.

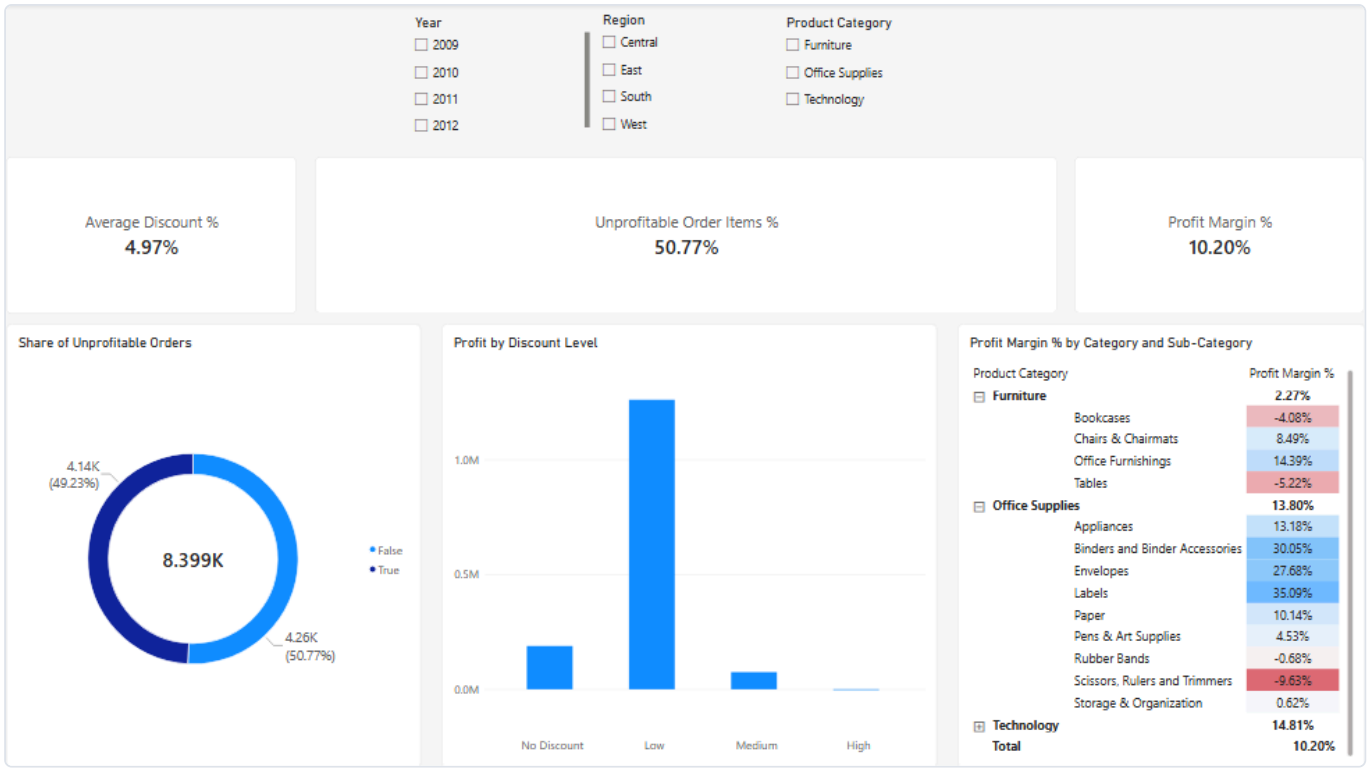
## 13 Power BI dashboard

THREE PAGES, CROSS-FILTERED BY YEAR, REGION AND CATEGORY

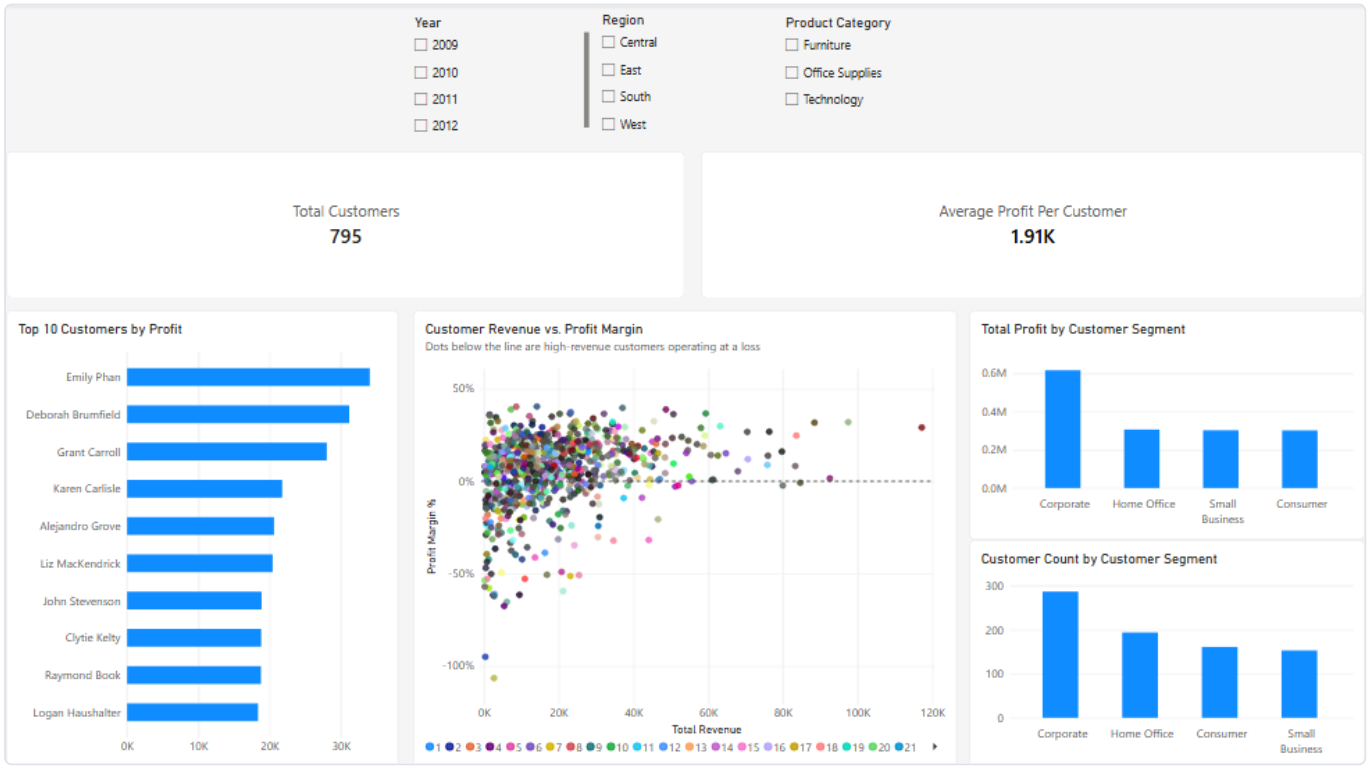
The dashboard is the delivery vehicle for the analysis — the artefact a manager opens rather than reads. Every page shares one slicer row, so any selection re-scopes all visuals simultaneously.




**Page 1 — Overview.** KPI header, then the revenue-and-profit trend. The two trend lines are drawn on one dollar axis, which is the point: profit is a thin band along the bottom of a much larger revenue line. The category pair at right shows revenue and margin ranking differently — Furniture is second on revenue and last on margin.




**Page 2 — Profitability.** The 50.77% unprofitable share is stated as a headline rather than buried. The category/sub-category matrix uses conditional formatting to make the four negative sub-categories — Bookcases, Tables, Rubber Bands, and Scissors, Rulers and Trimmers — findable at a glance, alongside the strong performers (Labels 35.09%, Binders 30.05%).




**Page 3 — Customer insights.** The scatter is the page's argument: total revenue on x, margin on y, with a reference line at zero. Every dot below that line is a customer spending real money at a loss — the §8 finding made visible in one view rather than derived from a table.

DESIGN DECISIONS WORTH NOTING

The dashboard deliberately leads with *margin* and *unprofitable share* alongside revenue on every page. Given that the central finding of the analysis is that revenue and profit disagree, a revenue-only dashboard would have reproduced the exact blind spot the study set out to expose.

# 14 Key findings

EIGHT RESULTS, WITH THE EVIDENCE EACH RESTS ON

## Thematic scorecard

Each of the seven themes from §2, judged on the evidence gathered against it.

| THEME | ASSESSMENT | HEADLINE MEASURE | WHAT DRIVES THE RATING |
|---|---|---|---|
| **Performance** | WATCH | 10.20% margin | Positive and stable in aggregate, but the trend within it is downward |
| **Geography** | SOUND | 1 of 48 states negative | Broadly healthy; West's margin gap is the one real concern |
| **Products** | ACTION | 2.27% Furniture margin | Concentrated, quantified losses in four sub-categories and ten SKUs |
| **Customers** | STRENGTH | 81.6% profit from repeat | Strong retention economics; unmanaged margin risk in a minority of accounts |
| **Pricing** | WATCH | 6.50% at medium discount | Not the cause of losses, but the Medium band shows a real 6.1pp degradation |
| **Operations** | SOUND | 2.03 days average | Uniform and reliable; a 1.25% delay tail and one open question on mode economics |
| **Time** | WATCH | 2012: +8.24% / −10.37% | No dependable seasonality; growth and profit move independently |

## Findings register

| # | FINDING | EVIDENCE | CONFIDENCE |
|---|---|---|---|
| 1 | **Revenue and profit are decoupled at every level of aggregation** | Rank inversions in region, segment, category, product, quarter and year. 2012: revenue +8.24%, profit −10.37%. | HIGH |
| 2 | **Furniture is the structural category weakness** | $5.18M revenue → $117K profit (2.27%), against Technology's 14.81% on comparable scale. | HIGH |
| 3 | **Losses concentrate in two sub-categories and ten products** | Tables −$99.1K and Bookcases −$33.6K are 94% of sub-category losses; ten high-revenue SKUs destroy $141.1K. | HIGH |
| 4 | **Discounting is not the driver of losses** | r = −0.037 with profit; 5.19% vs 4.74% average discount on loss-making vs profitable lines; 0.35pp spread across all sub-categories. | HIGH |
| 5 | **Repeat customers generate 82% of profit** | 467 frequent customers → $1.24M of $1.52M. A frequent buyer is worth 12.2× a one-time buyer. | HIGH |
| 6 | **Scale and efficiency diverge geographically** | South: smallest revenue, best margin (13.41%). West: second-largest revenue, worst margin (8.21%). | HIGH |
| 7 | **Fulfilment is uniform and is not a margin driver** | All modes and regions ship in 1.95–2.09 days; shipping_days correlates ≈0 with every financial variable. | HIGH |
| 8 | **Seasonality exists in aggregate but not year to year** | December strongest and June weakest when pooled; four different peak months across four years. | MODERATE |

SYNTHESIS

One sentence carries the study: **scale hides weak economics.** Every dimension examined contains at least one entity that looks strong on revenue and poor on profit — a region, a segment, a category, a product, a customer, a quarter. Any decision process that ranks by revenue alone will systematically select those entities for more investment.

# 15 Recommendations

ORDERED BY EXPECTED RETURN AGAINST EFFORT

| PRIORITY | ACTION | RATIONALE | AT STAKE |
|---|---|---|---|
| P1 | **Reprice or retire the ten loss-making high-revenue SKUs** | Demand is proven at current prices; the prices are below cost. This is ten line items, not a strategy change. | $141,149<br>9.3% of 4-yr profit |
| P1 | **Run a cost and pricing review on Tables and Bookcases** | −$132.6K across 550 order lines. Spread over that many transactions, this is structural, not anomalous. | $132,645<br>8.7% of 4-yr profit |
| P1 | **Report margin beside revenue in all management reporting** | The 2012 result was invisible in revenue reporting. Governance change, near-zero cost. | Prevents recurrence |
| P2 | **Introduce conditional discount approval above 10% on below-average-margin products** | Targets the Medium-band degradation (12.60% → 6.50%) without restricting the 82% of business priced sensibly. | $1.15M<br>revenue in scope |
| P2 | **Protect the 467 frequent accounts explicitly** | 82% of profit sits in this cohort. There is currently no retention mechanism distinguishing it. | $1,241,432 |
| P2 | **Investigate West's product mix and pricing** | 5.2 points behind South on margin. Shipping is ruled out; mix and pricing are the remaining explanations. | ≈$190K<br>if margin met South's |
| P2 | **Quantify the Delivery Truck margin gap against the Furniture problem** | 42% of revenue at a 4.33% margin. Establish how much is Furniture mix and how much is freight economics. | $6.22M<br>revenue in scope |
| P3 | **Targeted order review for Montana** | −$9,127 on $76,072. Single-state, so a local review is proportionate; no regional action warranted. | $9,127 |
| P3 | **Audit whether Express Air earns its premium** | 2.00 days against Regular Air's 2.04. If the cost differential is real, the service differential is not. | $1.18M<br>revenue in scope |

ON THE "AT STAKE" COLUMN

These figures are the **historical profit impact observed over 2009–2012**, not a forecast of recoverable value. Repricing a loss-making SKU will reduce its volume; the recoverable amount depends on price elasticity this dataset cannot measure. They are sizing estimates for prioritisation, not a business case.

**They must not be summed.** The first two P1 items overlap by $30,566, because five of the ten loss-making SKUs sit inside the Tables sub-category. Their combined exposure is **$251,130** — not $273,794.

## 16 Limitations

WHAT THIS ANALYSIS CANNOT TELL YOU

- **Association, not causation.** This is historical transactional data. It establishes that Furniture margins are thin; it cannot establish why without cost accounting the dataset does not contain.
- **Missing business drivers.** No marketing spend, customer acquisition cost, inventory position, supplier terms, competitor pricing or overhead allocation. The profit field is taken as given.
- **The High discount band is not analysable.** Two order lines and $1,002 of revenue cannot support any conclusion, and §9 declines to draw one.
- **Two margin definitions coexist.** The Python analysis uses revenue-weighted margin; parts of the SQL layer average line-level margins. They disagree by construction — see §12. Weighted margin should be preferred wherever a single figure is quoted.
- **Seasonality is unproven.** Four years, with a different peak month in each, is not enough to establish a recurring calendar.
- **Outliers materially move averages.** Sales skew is +5.39 and shipping days +14.25. Every average in this report should be read alongside the corresponding distribution or ranking; this is why the analysis reports both.
- **Grain sensitivity.** "50.77% of orders are unprofitable" is 50.77% of order *lines*. A whole order containing one loss-making line is not necessarily unprofitable.
- **Period.** The data ends December 2012. Findings describe that window and are not a statement about current trading.

> INTERPRETATION STANDARD
>
> Where evidence is associative the report says so. Where a finding rests on a small sample it says how small. Where two valid methods disagree it presents both and explains which question each answers. A portfolio analysis that only reports the findings that flatter it is not demonstrating analysis.

## 17 Appendix

SCOPE, REPRODUCIBILITY AND THE QUESTION SET

## The 39 business questions

| CATEGORY | QS | COVERS | REPORT § |
|---|---|---|---|
| **Overall business performance** | Q1–Q5 | Revenue, profit, margin, AOV, monthly trend, best and worst year | §5 |
| **Regional performance** | Q6–Q11 | Region revenue / profit / margin, state and city detail, loss-making states | §6 |
| **Product & category** | Q12–Q18 | Category and sub-category economics, top products, loss-making SKUs | §7 |
| **Customer analysis** | Q19–Q24 | Segment performance, top customers by revenue and profit, unprofitable accounts | §8 |
| **Discount & profitability** | Q25–Q29 | Correlation, discount by sub-category, band margins, deepest-discount orders | §9 |
| **Shipping & operations** | Q30–Q34 | Mode usage, delivery times, profit by mode, delay outliers, regional speed | §10 |
| **Time series & trends** | Q35–Q39 | Monthly and quarterly patterns, seasonality test, YoY growth | §11 |

## Metric definitions

Stated explicitly, because two of them are computed more than one way in the underlying work and the difference changes the answer.

| METRIC | DEFINITION USED IN THIS REPORT | NOTE |
|---|---|---|
| **Revenue** | `SUM(sales)` | Gross, as recorded in source; no returns field exists |
| **Profit** | `SUM(profit)` | Taken as given; the source does not decompose it into cost components |
| **Profit margin** | `SUM(profit) / SUM(sales) × 100` | **Revenue-weighted.** This is the definition used everywhere in §5–§11 |
| **Average line margin** | `AVG(profit_margin)` | Unweighted; treats a $6 line equally with a $60,000 line. Used only where §12 flags it |
| **Average order value** | `AVG(SUM(sales) GROUP BY order_id)` | Per *order*, not per line — $2,713.90 across 5,496 orders |
| **Order line** | One row — one product within one order | The analytical grain; 8,399 lines across 5,496 orders |
| **Unprofitable** | `profit < 0` on a line | 50.77% of lines. A line, not necessarily a whole order |
| **Discount** | Stored as a decimal (0.05 = 5%) | Range 0 – 0.25; mean 0.0497 |
| **Shipping days** | `ship_date − order_date`, in days | 0 is valid (same-day dispatch) |
| **Growth** | Period-over-period percentage change | Computed on totals, not on per-order averages |

### Data dictionary — 33 columns

23 source columns plus 10 engineered in Notebook 2.

| COLUMN | TYPE | ORIGIN | DESCRIPTION |
|---|---|---|---|
| `row_id` | int | Source | Unique line identifier, 1–8,399 |
| `order_id` | int | Source | Order identifier; 5,496 distinct values, repeats across lines |
| `order_date` | datetime | Source | Date the order was placed; 1,418 distinct dates |
| `ship_date` | datetime | Source | Date shipped; never earlier than `order_date` (verified) |
| `order_priority` | text | Source | Critical / High / Medium / Low / Not Specified — near-even split |
| `order_quantity` | int | Source | Units on the line, 1–50; total 214,777 |
| `sales` | float | Source | Line revenue, $2.24 – $89,061.05; skew +5.39 |
| `discount` | float | Source | Decimal fraction, 0 – 0.25 |
| `profit` | float | Source | Line profit, –$14,140.70 – $27,220.69 |
| `unit_price` | float | Source | $0.99 – $6,783.02; the most skewed field (+14.13) |
| `shipping_cost` | float | Source | $0.49 – $164.73, mean $12.84 |
| `customer_name` | text | Source | 795 distinct customers |
| `city` | text | Source | 1,421 distinct cities |
| `state` | text | Source | 48 states |
| `zip_code` | int | Source | 1,626 distinct; staged as text in SQL to preserve leading zeros |
| `region` | text | Source | Central / East / South / West |
| `customer_segment` | text | Source | Corporate / Home Office / Consumer / Small Business |
| `product_category` | text | Source | Technology / Furniture / Office Supplies |
| `product_sub_category` | text | Source | 17 sub-categories |
| `product_name` | text | Source | 1,263 distinct products |
| `product_container` | text | Source | 7 packaging types |
| `product_base_margin` | float | Source | 0.35 – 0.85; bimodal (~0.38 and ~0.58). **63 values imputed** |
| `ship_mode` | text | Source | Regular Air (74.65%) / Delivery Truck / Express Air |
| `shipping_days` | int | Engineered | 0 – 92 days, mean 2.03; kurtosis 479 |
| `profit_margin` | float | Engineered | Line-level percentage, –1,144% – 97.73% |
| `year` | int | Engineered | 2009 – 2012 |
| `month` | int | Engineered | 1 – 12 |
| `month_name` | text | Engineered | Readable label for chart axes |
| `quarter` | text | Engineered | Q1 – Q4 |
| `day_of_week` | text | Engineered | Monday – Sunday; distribution near-even |
| `is_profitable` | bool | Engineered | `profit > 0`; True 49.23%, False 50.77% |
| `discount_band` | category | Engineered | Ordered: No Discount / Low / Medium / High |

| COLUMN | TYPE | ORIGIN | DESCRIPTION |
|---|---|---|---|
| `sales_band` | category | Engineered | Ordered: Small / Medium / Large / Very Large |

## Reproducibility

| ARTEFACT | PATH | CONTENTS |
|---|---|---|
| **Cleaned dataset** | `1_data/cleaned/superstore_cleaned.csv` | 8,399 × 33 — the single source for every figure in this report |
| **Notebook 1** | `2_notebooks/01_data_understanding.ipynb` | Structure, dtypes, missing values, cardinality, financial overview |
| **Notebook 2** | `2_notebooks/02_data_cleaning.ipynb` | Imputation, negative-profit investigation, 10 engineered features, validation |
| **Notebook 3** | `2_notebooks/03_eda.ipynb` | Distributions, IQR outliers, correlation matrix, profitability patterns |
| **Notebook 4** | `2_notebooks/04_Business_Analysis.ipynb` | All 39 questions with charts and written interpretation |
| **Schema** | `3_sql/schema.sql` | Staging table, four normalised tables, population and integrity checks |
| **Analysis queries** | `3_sql/analysis_queries.sql` | Ten advanced queries with inline findings |
| **Dashboard** | `4_dashboard/superstore_dashboard.pbix` | Three cross-filtered pages |

## Environment

Python 3 with pandas, NumPy, Matplotlib and Seaborn, developed in Google Colab. PostgreSQL queried through DBeaver. Power BI Desktop for the dashboard. All figures in this report were recomputed directly from `superstore_cleaned.csv` and reconcile to the notebook outputs.

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Abdullah Waseem

Python · Pandas · NumPy · Matplotlib · Seaborn · PostgreSQL · Power BI


Superstore Performance Review — an end-to-end analysis of 8,399 order lines covering January 2009 to December 2012. Every figure is reproducible from the cleaned dataset and the four notebooks in the project repository. September 2026.